

### Contact

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### Research Interests

Unstable and Stable (Motivic) Homotopy Theory, Synthetic Deformations, Algebraic  $K$ -theory, Algebraic Geometry in positive characteristic.

### Education

- 2023–present **PhD in Mathematics, JGU Mainz**  
Thesis: *Localization functors in unstable motivic homotopy theory*  
Advisor: Tom Bachmann
- 2022 **MSc in Mathematics (grade: 1.0), LMU Munich**  
Thesis: *Isomotives of Dimension at most 1*  
Advisor: Fabien Morel
- 2021 **BSc in Mathematics, LMU Munich**  
Thesis: *Examples of étale  $(\varphi, \Gamma)$ -modules*  
Advisor: Werner Bley
- 2017–2022 **Studies in Mathematics, LMU Munich**

### Publications

- 2026 **The derived  $\infty$ -category of Frobenius Modules**, (with Timo Weiß), to appear: *Journal of Algebra*, [arXiv:2510.23267](https://arxiv.org/abs/2510.23267)
- 2026 **Unstable arithmetic fracture squares in  $\infty$ -topoi**, *Journal of Homotopy and Related Structures*, [arXiv:2404.18618](https://arxiv.org/abs/2404.18618)
- 2025 **The derived  $\infty$ -category of Cartier Modules**, (with Timo Weiß), *Journal of Pure and Applied Algebra*, [arXiv:2410.17102](https://arxiv.org/abs/2410.17102)

### Preprints

- 2025 **Nilpotence of  $\eta$  in étale motivic spectra**, (with Swann Tubach), submitted, [arXiv:2511.09476](https://arxiv.org/abs/2511.09476)
- 2025 **Monadic resolutions for generalized spaces**, (with Tom Bachmann and Anton Engelman), preprint, [arXiv:2511.07064](https://arxiv.org/abs/2511.07064)
- 2025 **Unstable étale motives**, submitted, [arXiv:2507.20320](https://arxiv.org/abs/2507.20320)
- 2024 **The pro-Nisnevich topology**, submitted, [arXiv:2404.17314](https://arxiv.org/abs/2404.17314)
- 2024 **Unstable  $p$ -completion in motivic homotopy theory**, submitted, [arXiv:2401.17848](https://arxiv.org/abs/2401.17848)

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## Selected Talks

- June 2026 **Étale synthetic deformations**, Young Topologists Meeting, Copenhagen
- May 2026 **Monadic resolutions for generalized spaces**, Universität Regensburg
- Mar 2026 **Cdh-local motivic cohomology**, Recent Advances in Motivic Cohomology, Osnabrück
- Feb 2026 **Monadic resolutions for generalized spaces**, Universität Wuppertal
- Jan 2026 **Monadic resolutions for generalized spaces**, University of Copenhagen
- July 2025 **Étale rigidity for motivic spaces**, Motives and Arithmetic Geometry, Darmstadt
- June 2025 **Canonical resolutions for motivic spaces**, Young Topologists Meeting, Stockholm
- April 2025 **Étale rigidity for motivic spaces**, École Normale Supérieure, Lyon
- April 2025 **Étale rigidity for motivic spaces**, University of Toronto
- June 2024 **Proof of the Hopkins-Morel-Hoyois theorem**, International Workshop on Algebraic Topology, Shanghai
- Feb 2024 **Unstable  $p$ -completion in motivic homotopy theory**, YoungHom Seminar, online

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## Teaching Experience

- Summer 24 **Lecturer (1/3 of course)**, *Algebraic Topology I*, JGU Mainz  
*Teaching Assistant / Tutor*
- Summer 26 **Teaching Assistant**, *Math for Computer Science 2a*, JGU Mainz
- Winter 25/26 **Teaching Assistant**, *Math for Computer Science 1*, JGU Mainz
- Winter 24/25 **Teaching Assistant**, *Math for Computer Science 1*, JGU Mainz
- Winter 23/24 **Teaching Assistant**, *Math for Computer Science 2b*, JGU Mainz
- Summer 23 **Teaching Assistant**, *Differential Geometry*, JGU Mainz
- Summer 21 **Tutor**, *Commutative Algebra*, LMU Munich
- Winter 20/21 **Tutor**, *Algebra*, LMU Munich
- Summer 20 **Tutor**, *Linear Algebra 2*, LMU Munich
- Winter 19/20 **Tutor**, *Linear Algebra 1*, LMU Munich

## Organization

- 2023–2026 **Organizer**, *PhD-Seminar*, JGU Mainz
- Winter 26 **Organizer**, *Seminar on algebraic K-theory and chromatic homotopy theory*, JGU Mainz
- Summer 25 **Organizer**, *Seminar on motivic spectra*, JGU Mainz
- Summer 20 **Organizer**, *Reading Class on Category Theory*, LMU Munich

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## Academic Service

- Oct 2026 **Organizer**, *Workshop on “Motivic Loop Spaces”*, JGU Mainz
- Oct 2025 **Organizer**, *GAUS Workshop on “Motives and Higher Categories”*, JGU Mainz
- Jan 2025 **Organizer**, *GAUS Junior AG: “Maps between spherical group rings”*, JGU Mainz
- Mar 2024 **Co-organizer**, *Winter school on unstable motivic homotopy theory*, JGU Mainz

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## Non-academic Service

- 2016–present **Volunteer**, *Red Cross (BRK)*, Munich
- 2016–2023 **Software Developer**, *Microstep AG*

2019–2020 **Volunteer Author**, *Serlo*, Open-source platform for higher mathematics education

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## Languages

German Native

English Fluent